

1st Part Problem Set for Applied Abstract Algebra

Quivers: representations and morphisms

1. Let $M, M' \in \text{rep } Q$. Show that the set of morphism $\text{Hom}(M, M')$ is a k -vector space.
2. Let Q be the quiver $1 \longrightarrow 2 \longleftarrow 3$ and consider the representations

$$M \quad k \xrightarrow{\begin{bmatrix} 1 \\ 0 \end{bmatrix}} k^2 \xleftarrow{\begin{bmatrix} 0 \\ 1 \end{bmatrix}} k$$

$$M' \quad k \xrightarrow{\begin{bmatrix} 1 \\ 0 \end{bmatrix}} k^2 \xleftarrow{\begin{bmatrix} 1 \\ 0 \end{bmatrix}} k$$

- (a) Show that M and M' are not indecomposable.
 - (b) Show that M and M' are not isomorphic.
3. Let Q be the quiver $1 \longrightarrow 2 \longleftarrow 3$ and let M be a representation

$$k^2 \xrightarrow{\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}} k^3 \xleftarrow{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}} k .$$

- (a) Write M as a direct sum of the indecomposable representations listed below:

$$S(1) : k \longrightarrow 0 \longleftarrow 0 \quad S(2) : 0 \longrightarrow k \longleftarrow 0 \quad S(3) : 0 \longrightarrow 0 \longleftarrow k$$

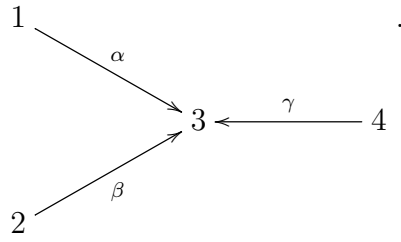
$$P(1) : k \xrightarrow{1} k \longleftarrow 0 \quad I(2) : k \xrightarrow{1} k \xleftarrow{1} k \quad P(3) : 0 \longrightarrow k \xleftarrow{1} k$$

- (b) Show that there is a non-split short exact sequence

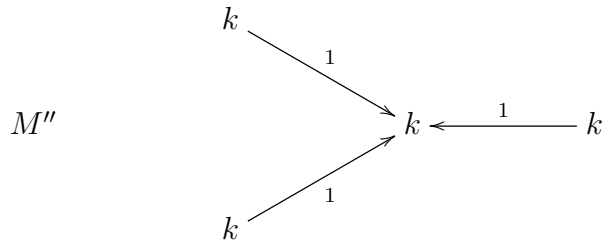
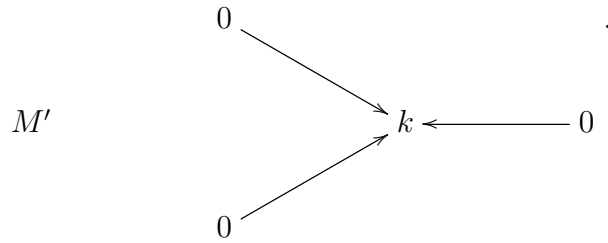
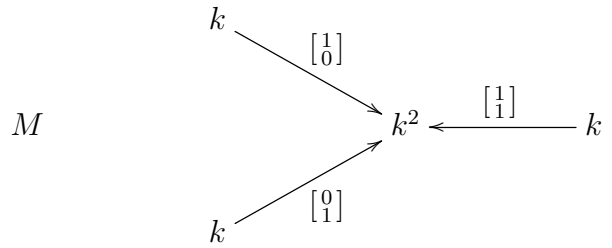
$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0 .$$

4. Find all indecomposable representations up to isomorphism of the quiver $1 \longrightarrow 2$.

5. Let Q be the quiver



and consider the following representations



(a) Show that

$$\begin{aligned} \operatorname{Hom}(M, M') &= 0, & \operatorname{Hom}(M', M) &\simeq k^2 \\ \operatorname{Hom}(M, M'') &\simeq k^2, & \operatorname{Hom}(M'', M) &= 0. \end{aligned}$$

(b) Show that M is not isomorphic to $M' \oplus M''$.

(c) Show that there is a short exact sequence:

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0.$$

6. Let Q be the quiver

$$1 \begin{array}{c} \xleftarrow{\alpha} \\ \xleftarrow{\beta} \end{array} 2$$

For any $\lambda \in k \cup \{\infty\}$, define M_λ to be the representation:

$$1 \begin{array}{c} \xleftarrow{1} \\ \xleftarrow{\lambda} \end{array} 2 \quad \lambda \in k;$$

$$1 \begin{array}{c} \xleftarrow{0} \\ \xleftarrow{1} \end{array} 2 \quad \lambda = \infty.$$

- (a) Show that each M_λ is indecomposable.
- (b) Show that $M_\lambda \simeq M_\mu$ if and only if $\lambda = \mu$. In particular, the number of indecomposable representations depends on the choice of the field k .
- (c) Show that $\text{Hom}(M_\lambda, M_\mu) = 0$ if $\lambda \neq \mu$.
- (d) Show that for each λ there is a short exact sequence

$$0 \longrightarrow S(1) \longrightarrow M_\lambda \longrightarrow S(2) \longrightarrow 0 ,$$

where 1 and 2 are the representations

$$S(1) : k \begin{array}{c} \xleftarrow{\quad} \\ \xleftarrow{\quad} \end{array} 0 \quad S(2) : 0 \begin{array}{c} \xleftarrow{\quad} \\ \xleftarrow{\quad} \end{array} k .$$

- 7. Let $f : L \rightarrow M$ be a morphism of representations. Prove that the definition of $\text{coker } f$ is equivalent to the one defined in classroom through universal property (otherwise, see Remark 1.6 in Schiffler's book).
- 8. Let M , M' and N be representations of Q and let $f : M \rightarrow N$, $g : M' \rightarrow N$ be morphisms. Define the fiber product (or pull back) of f and g as

$$X = \{(a, b) \mid a \in M, b \in M', \text{ such that } f(a) = g(b)\},$$

and define the projections $\pi_1(a, b) = a$ and $\pi_2(a, b) = b$.

- (a) Show that X is a subrepresentation of $M \oplus M'$ and that there is a commutative diagram:

$$\begin{array}{ccc} X & \xrightarrow{\pi_2} & M' \\ \pi_1 \downarrow & & \downarrow g \\ M & \xrightarrow{f} & N \end{array}$$

- (b) Show that if f is injective then π_2 is injective.
- (c) Show that if f is surjective then π_2 is surjective.

- (d) Suppose $0 \longrightarrow L \xrightarrow{h} M \xrightarrow{f} N \longrightarrow 0$ is a short exact sequence, define $h' : L \rightarrow X$ by $h'(n) = (h(n), 0)$. Show that the following diagram is commutative with exact rows:

$$\begin{array}{ccccccccc}
 0 & \longrightarrow & L & \xrightarrow{h'} & X & \xrightarrow{\pi_2} & M' & \longrightarrow & 0 \\
 & & \downarrow 1_L & & \downarrow \pi_1 & & \downarrow g & & \\
 0 & \longrightarrow & L & \xrightarrow{h} & M & \xrightarrow{f} & N & \longrightarrow & 0
 \end{array}$$

9. Let $f : L \rightarrow M$ be a morphism. Then a cokernel of f is a morphism $g : M \rightarrow N$ such that $gf = 0$, and given any morphism $v : M \rightarrow X$ such that $vf = 0$, there is a unique morphism $u : N \rightarrow X$ such that $ug = v$.

$$\begin{array}{ccccc}
 L & \xrightarrow{f} & M & \xrightarrow{g} & N \\
 & & & \searrow v & \downarrow \exists u \\
 & & & & X
 \end{array}$$

Prove that the two definitions of cokernel agree.

10. Let Q be a quiver and $L \xrightarrow{f} M \xrightarrow{g} N \longrightarrow 0$ a sequence in $\text{rep } Q$. Show that this sequence is exact if and only if for every representation $X \in \text{rep } Q$, the sequence

$$0 \longrightarrow \text{Hom}(N, X) \xrightarrow{g^*} \text{Hom}(M, X) \xrightarrow{f^*} \text{Hom}(L, X)$$

is exact.

11. Let $g : M \rightarrow N$ be a surjective morphism between representations of Q , and let $P(i)$ be the projective representation at vertex i . Show that the map

$$g_* : \text{Hom}(P(i), M) \rightarrow \text{Hom}(P(i), N)$$

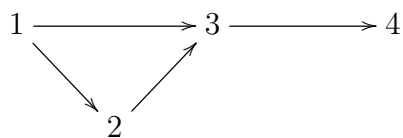
is surjective.

12. Let $g : L \rightarrow M$ be an injective morphism between representations of Q , and let $I(i)$ be the injective representation at vertex i . Show that the map

$$g^* : \text{Hom}(M, I(i)) \rightarrow \text{Hom}(L, I(i))$$

is surjective.

13. Let Q be the quiver



Prove that

$$P(1) \simeq k \begin{array}{c} \xrightarrow{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}} k^2 \\ \searrow z \quad \nearrow \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \\ \quad k \end{array} \xrightarrow{\begin{bmatrix} a & b \\ c & d \end{bmatrix}} k^2,$$

if and only if all four maps have maximal rank, that is, $ad - bc \neq 0$, $(x_1, x_2) \neq (0, 0)$, $(y_1, y_2) \neq (0, 0)$, $z \neq 0$ and the vectors

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

are linearly independent.

14. Compute the indecomposable projective (resp. injective) representations $P(i)$ (resp. $I(i)$) for the following quivers:

$$Q_1 : \quad 1 \rightleftarrows 2 \rightleftarrows 3$$

$$Q_2 : \quad \begin{array}{ccccc} 1 & \xrightarrow{\quad} & 2 & \xrightarrow{\quad} & 3 \\ & & \downarrow & & \nearrow \\ & & 4 & & \end{array}$$

15. Compute a projective resolution for each simple representation $S(i)$ for quivers Q_1 and Q_2 from exercise 14.